

USB Communication



File: usb.kicad_sch

Power Supply



File: power_supply.kicad_sch

MCU



File: mcu.kicad_sch

Heating Element



File: heating_element.kicad_sch

Cooling System



File: cooling_system.kicad_sch

PCB Mounting Holes

- H1

MountingHole
- H2

MountingHole
- H3

MountingHole
- H4

MountingHole

PCB Fiducials

- FID1

Fiducial
- FID2

Fiducial
- FID3

Fiducial
- FID4

Fiducial
- FID5

Fiducial
- FID6

Fiducial
- FID7

Fiducial
- FID8

Fiducial

SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /
File: atech_reflow_plate_v2.kicad_sch

Title: ReflowDesk AT-MK1

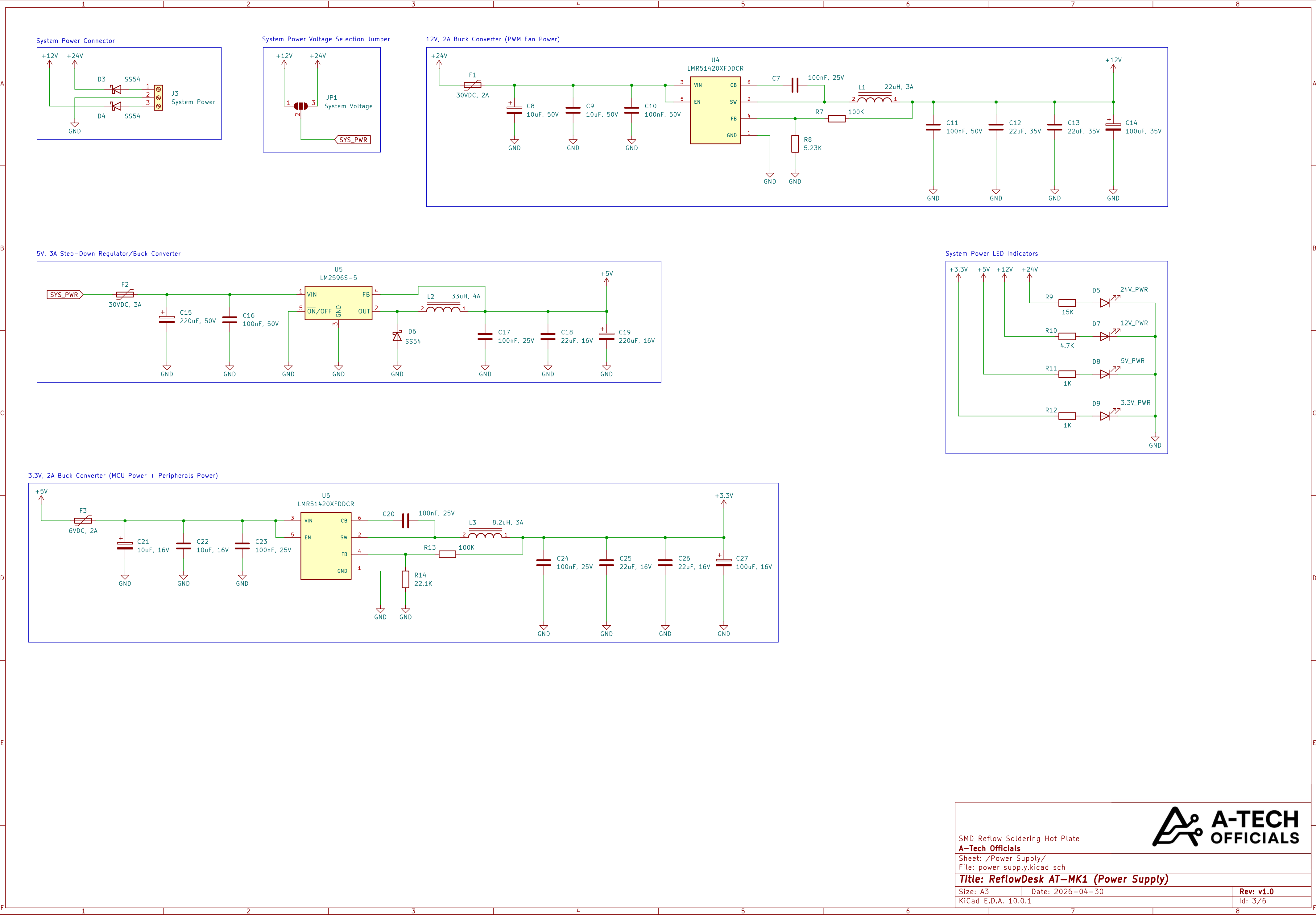
Size: A3
KiCad E.D.A. 10.0.1

Date: 2026-04-30

Rev: v1.0

Id: 1/6





SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /Power Supply/

File: power_supply.kicad_sch

Title: ReflowDesk AT-MK1 (Power Supply)

Size: A3

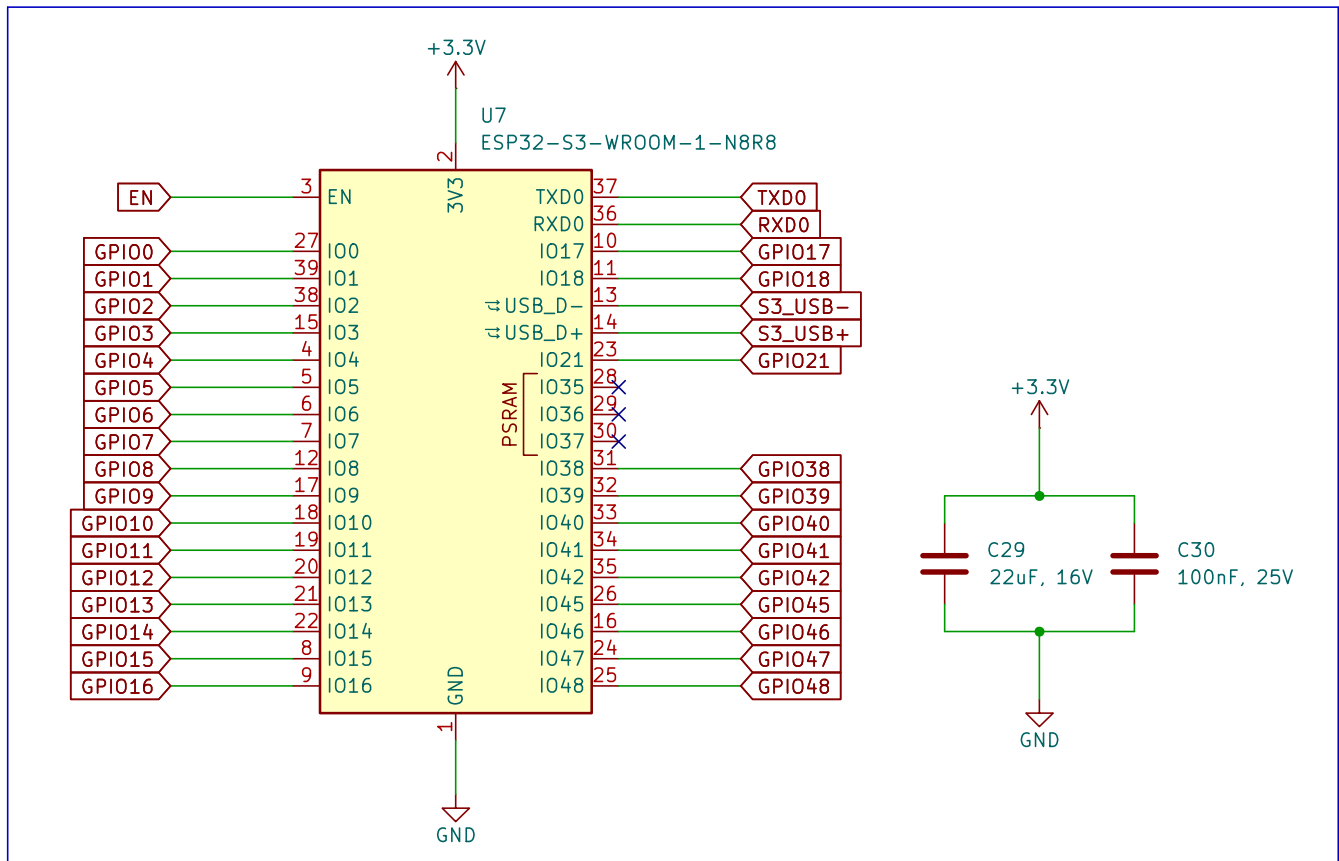
Date: 2026-04-30

Rev: v1.0

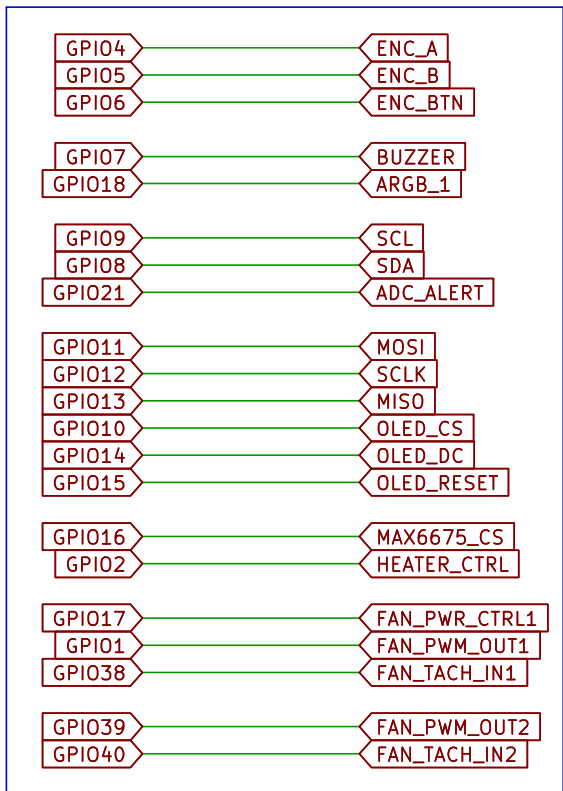
KiCad E.D.A. 10.0.1

Id: 3/6

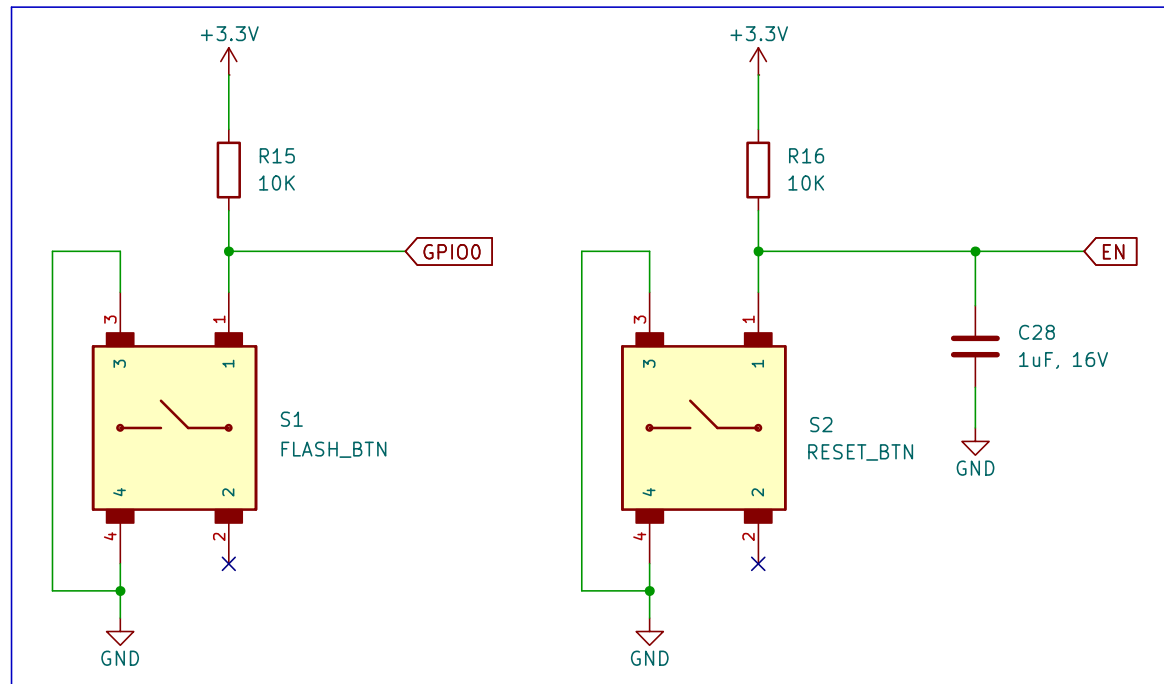
32-bit MCU (Flash: 8MB, PSRAM: 8MB)



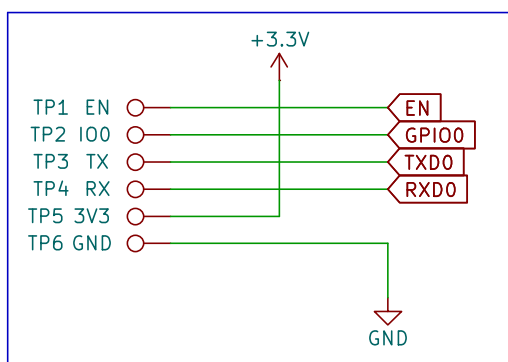
MCU GPIO Pins Allocation/Configuration



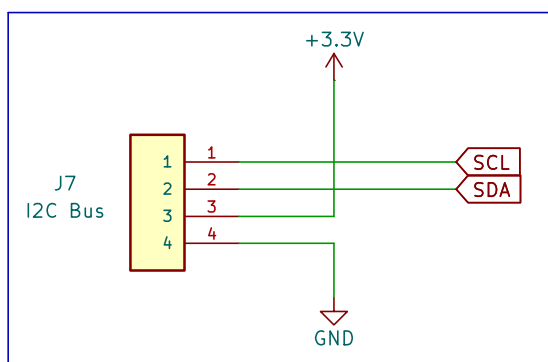
MCU Flash and Reset Buttons



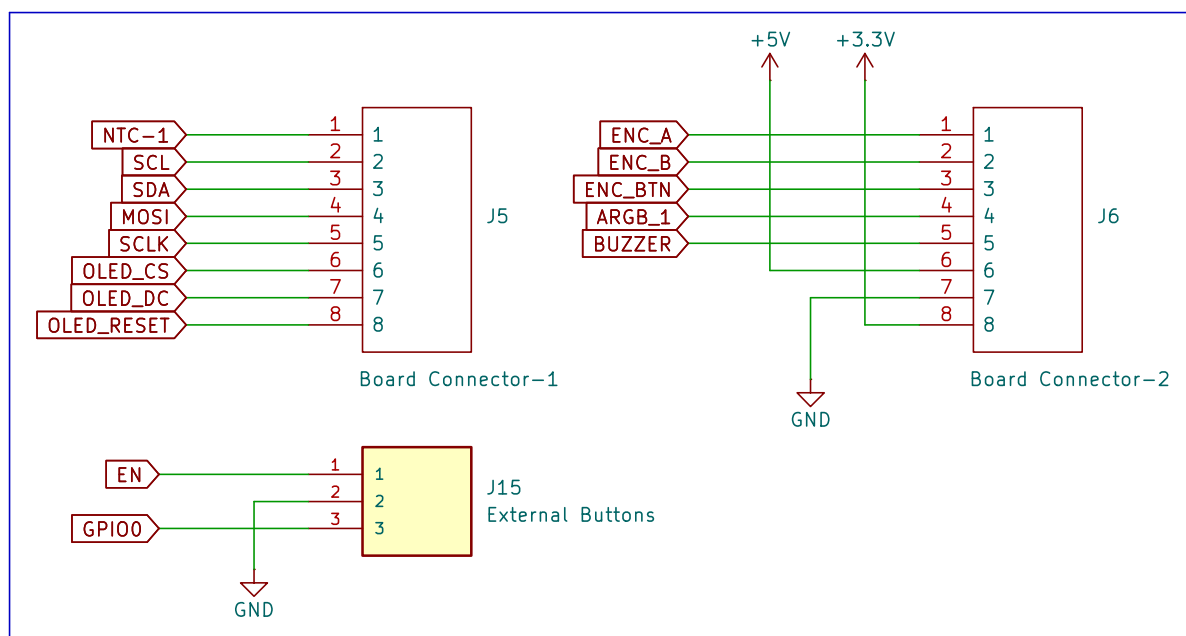
UART Test Pads



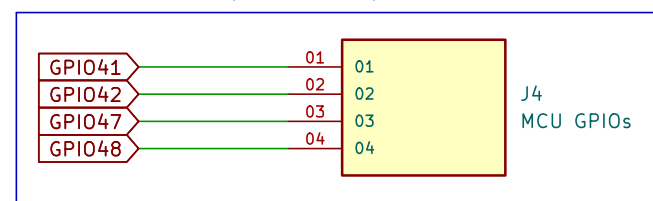
I2C Bus Connector



DaughterBoard (User Interface Board) Connectors



MCU GPIO Breakout (Future Usage)



SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /MCU/

File: mcu.kicad_sch

Title: ReflowDesk AT-MK1 (MCU)

Size: A3

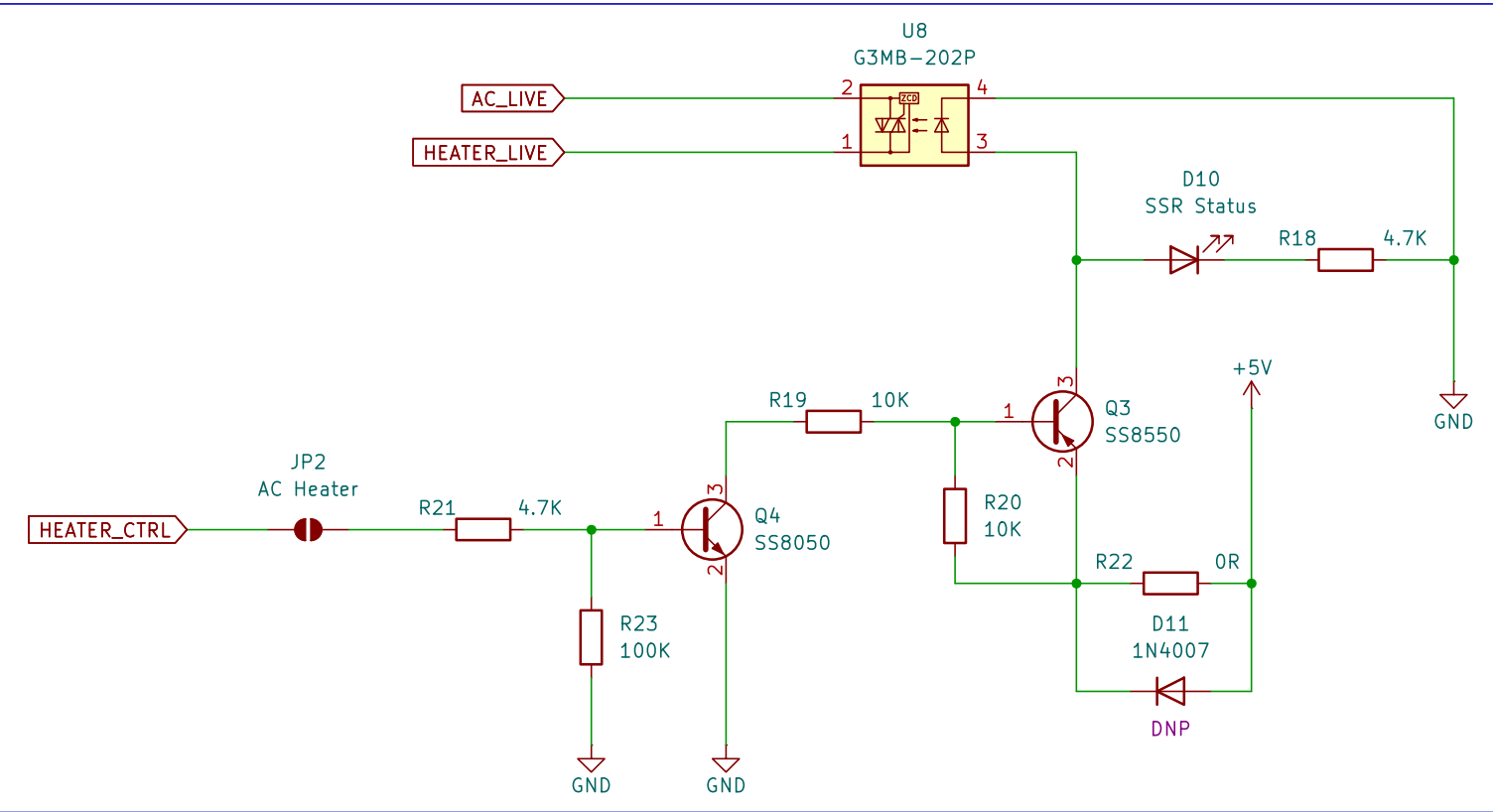
Date: 2026-04-30

Rev: v1.0

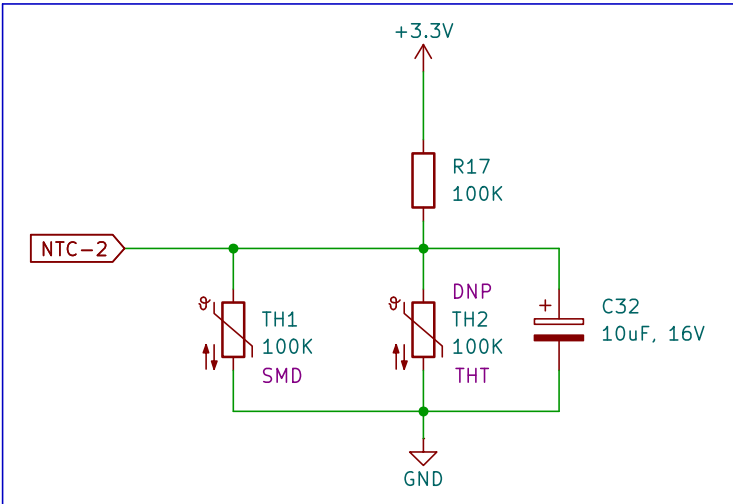
KiCad E.D.A. 10.0.1

Id: 4/6

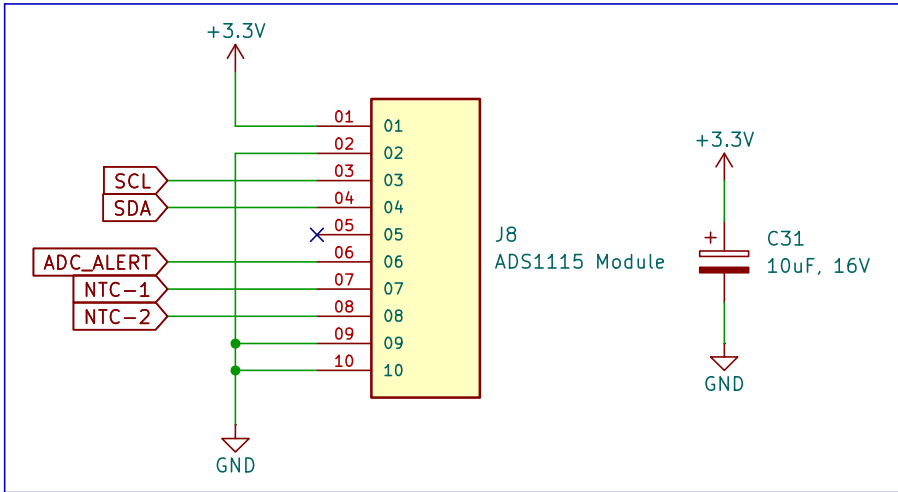
220VAC PTC Heating Element Control (Active-Low SSR, Active-High Signal)



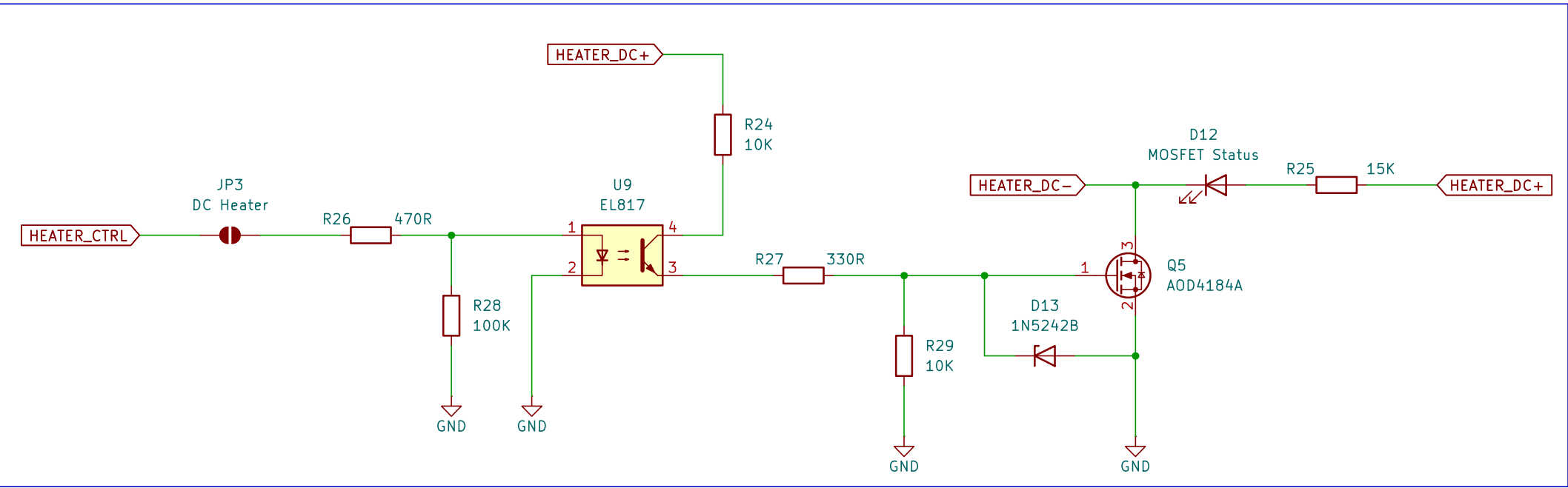
NTC Ambient Temp Sensor (Controller Board)



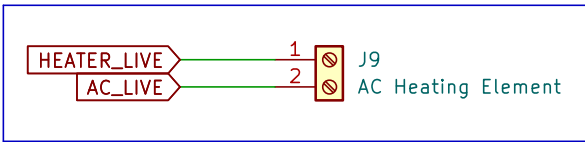
ADS1115 16-bit ADC Module Connector



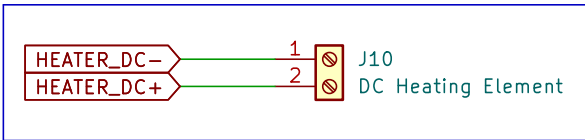
DC PTC Heating Element Control (Low-Side MOSFET Switch)



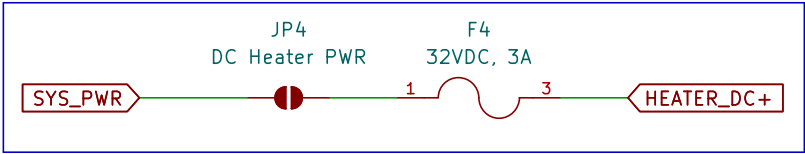
220VAC PTC Heating Element Connector



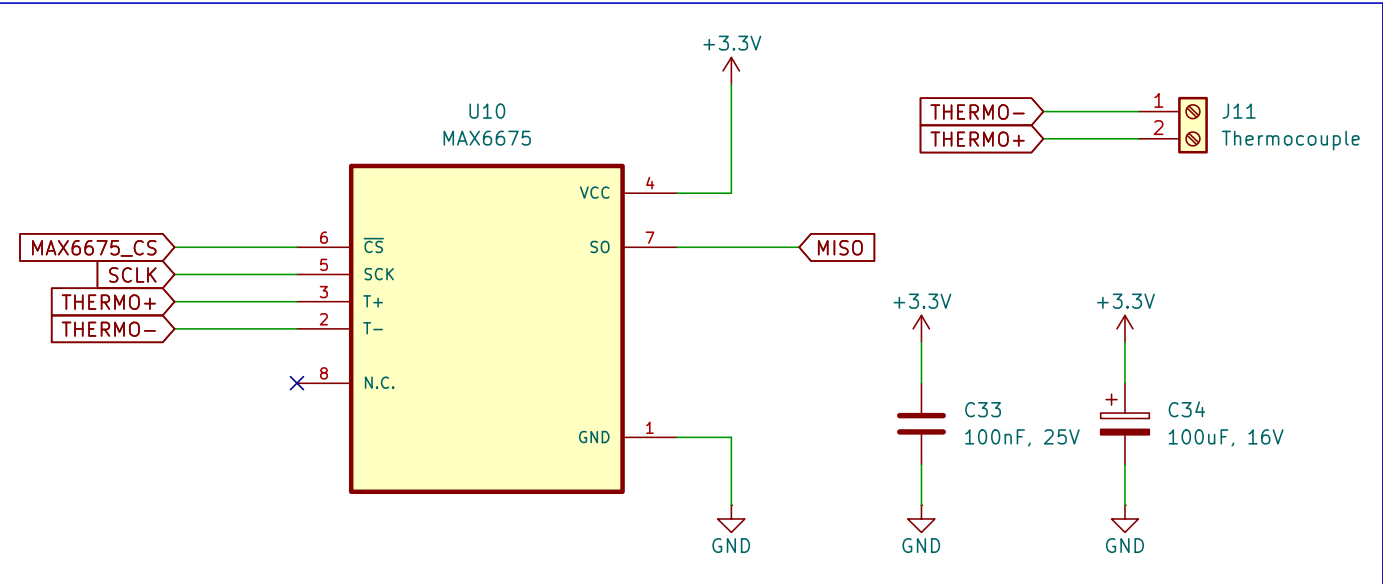
DC PTC Heating Element Connector



DC PTC Heating Element Power Enable Jumper



Hot Plate Temperature Sensor (K-Type Thermocouple)



SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /Heating Element/

File: heating_element.kicad_sch

Title: ReflowDesk AT-MK1 (Heating Element)

Size: A3

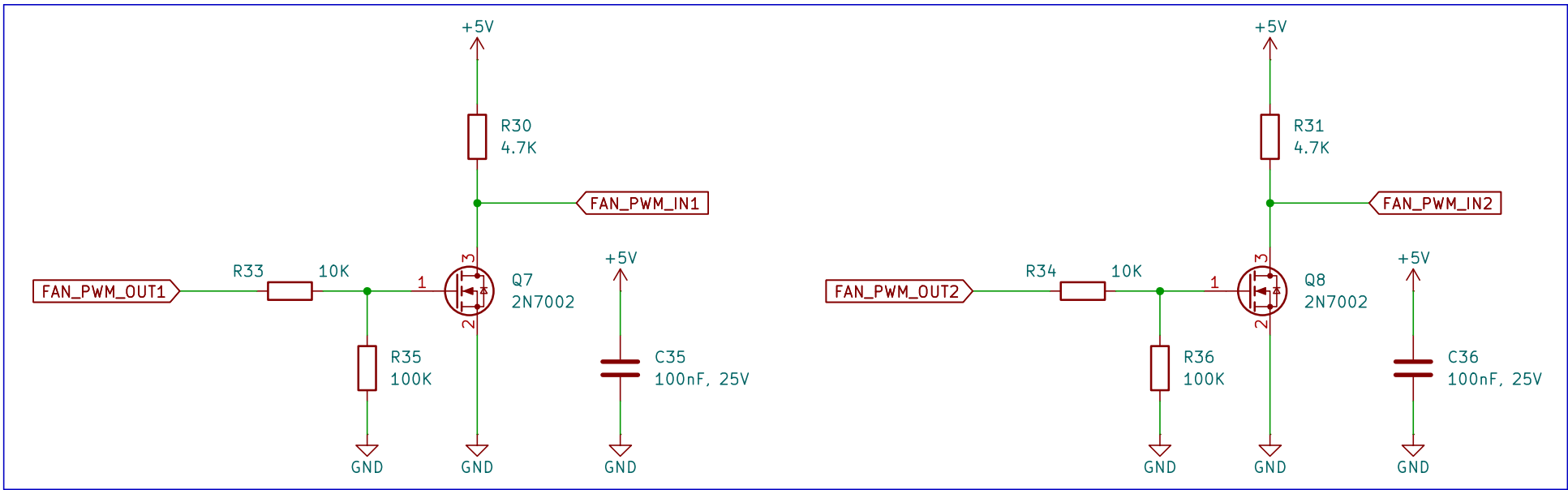
Date: 2026-04-30

Rev: v1.0

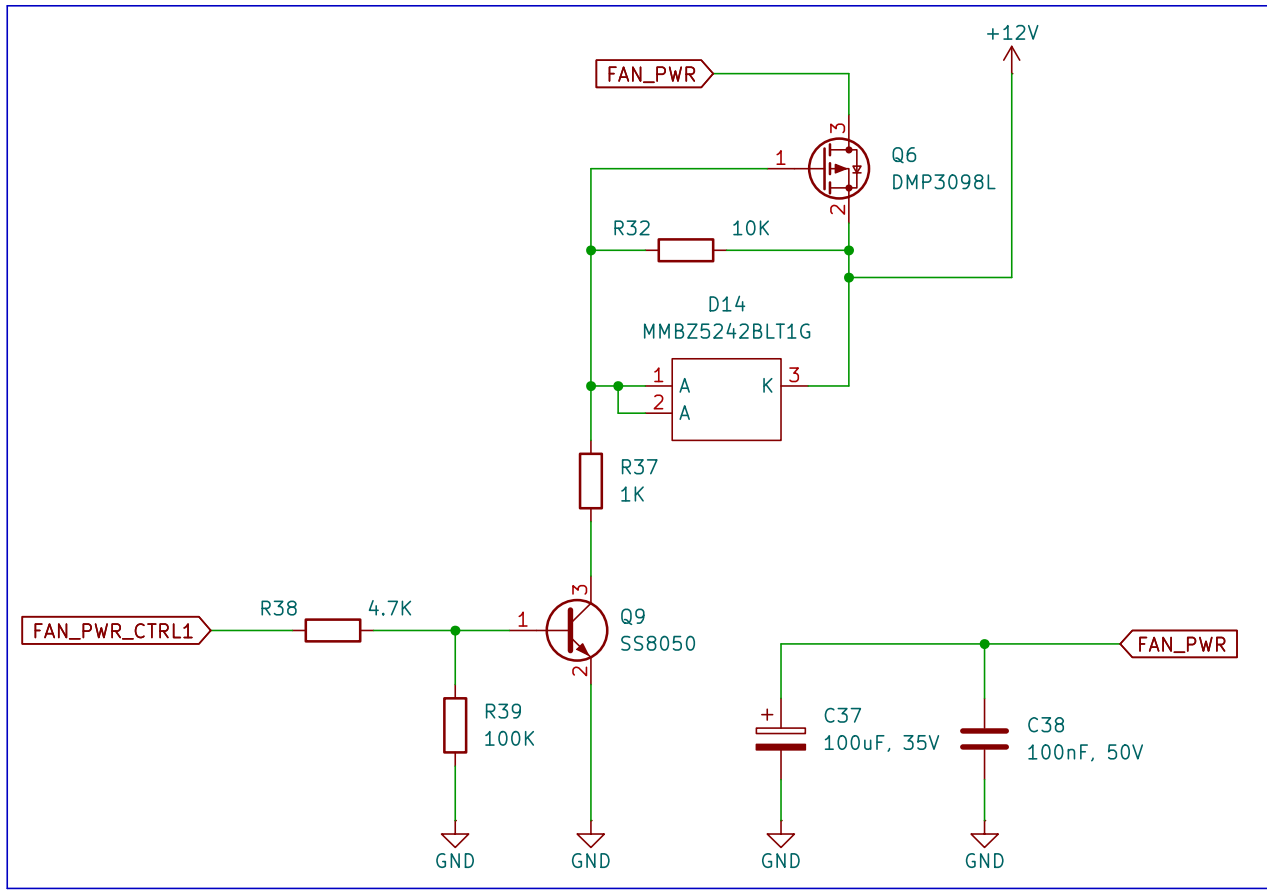
KiCad E.D.A. 10.0.1

Id: 6/6

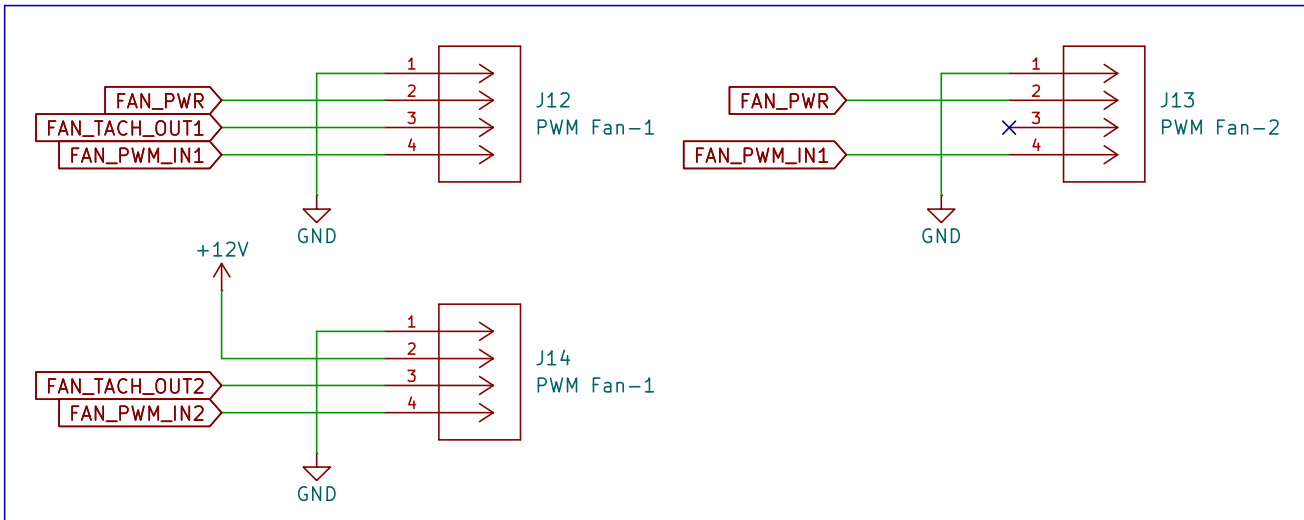
N-MOSFET Open-Drain Level Shifter / PWM Distribution Buffer



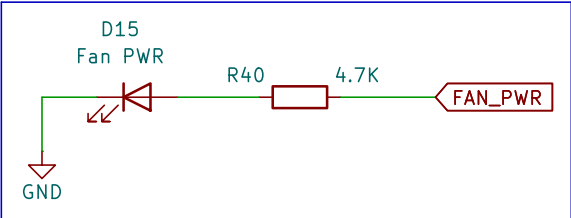
12VDC PWM Fan Power Control



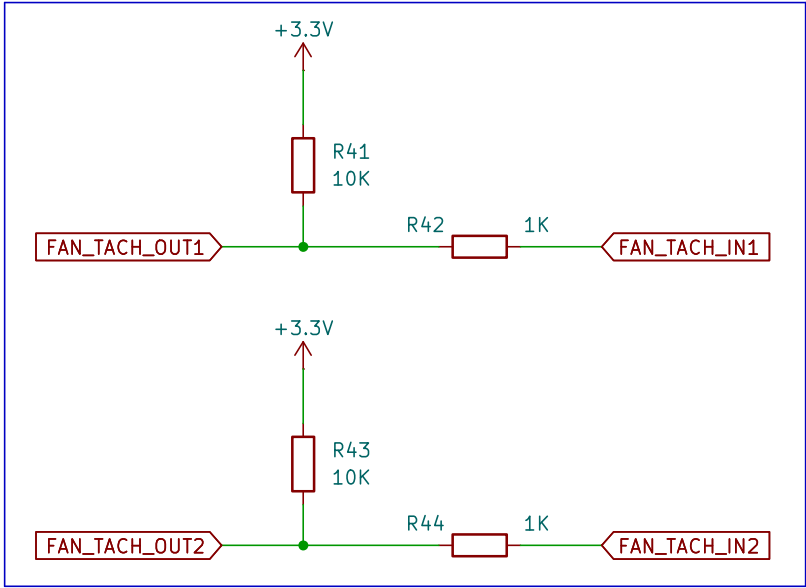
12VDC PWM Fan Connectors



Fan Power Status LED



Fan Tachometer Read Circuit



SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /Cooling System/

File: cooling_system.kicad_sch

Title: ReflowDesk AT-MK1 (Cooling System)

Size: A3

Date: 2026-04-30

Rev: v1.0

KiCad E.D.A. 10.0.1

Id: 7/6